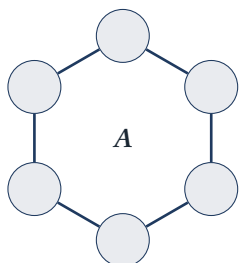


COMPANION VOLUME TO A CLASSROOM TUTORIAL

# The Mathematics of Message Passing, with Chemical Examples

*Theorems with proofs, and twelve worked problems from chemistry*



eigenvalues 2, 1, 1, -1, -1, -2

$$E_{\pi} = 6\alpha + 8\beta$$

the adjacency matrix a graph model propagates  
with is the Hückel matrix

*Part A states and proves the results that a message-passing model relies on: walk counting, permutation equivariance, injectivity of the aggregator on multisets, the spectral derivation of graph convolution, the contraction that causes oversmoothing, and the Euclidean symmetry required of a potential. Part B applies each of them to a concrete chemical question, with the full solution worked out. Every numerical value quoted was computed by the script reproduced in Appendix A.*

Intended for one problem-class session per part

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# Part A. The mathematics

## 1 Matrices attached to a graph

Let  $\mathcal{G}$  have  $n$  nodes and let  $A$  be its adjacency matrix,  $D$  the diagonal degree matrix  $D_{ii} = d_i = \sum_j A_{ij}$ .

### The graph Laplacian

$$L = D - A, \quad L_{\text{sym}} = D^{-1/2} L D^{-1/2} = I - D^{-1/2} A D^{-1/2}. \quad (1)$$

**Proposition 1.** For any  $f \in \mathbb{R}^n$ ,

$$f^\top L f = \frac{1}{2} \sum_{i,j} A_{ij} (f_i - f_j)^2 \geq 0. \quad (2)$$

Consequently  $L$  is positive semidefinite, its smallest eigenvalue is zero with eigenvector  $\mathbf{1}$ , and the multiplicity of the zero eigenvalue equals the number of connected components. The eigenvalues of  $L_{\text{sym}}$  lie in  $[0, 2]$ .

*Proof.* Expand the right side of (2):  $\frac{1}{2} \sum_{i,j} A_{ij} (f_i^2 - 2f_i f_j + f_j^2) = \sum_i d_i f_i^2 - \sum_{i,j} A_{ij} f_i f_j = f^\top D f - f^\top A f$ . Non-negativity is immediate. Setting  $f = \mathbf{1}$  gives zero, and conversely a vanishing quadratic form forces  $f_i = f_j$  across every edge, so  $f$  is constant on each component. The upper bound follows from  $g^\top (I + D^{-1/2} A D^{-1/2}) g \geq 0$ , which is (2) with a plus sign in place of the minus. ■

The quantity in (2) is the *Dirichlet energy* of the signal  $f$  on the graph, and it measures how much  $f$  varies between bonded atoms. Section 6 shows that message passing drives it to zero, which is exactly the statement that all atoms end up looking alike.

### Chemist's note: three familiar matrices, one object

For a conjugated hydrocarbon in the Hückel approximation the effective Hamiltonian is  $H = \alpha I + \beta A$ , so the adjacency matrix that a graph model multiplies by at every layer is, up to the affine map  $x \mapsto \alpha + \beta x$ , the Hückel Hamiltonian. The Laplacian plays the same role in the harmonic problem, where  $L$  with unit force constants is the mass-weighted Hessian of a bead-and-spring model. A graph model propagating information along bonds is doing arithmetic a chemist has done since 1931.

## 2 Walks, powers and the receptive field

**Theorem 2** (Walk counting). For every integer  $k \geq 0$ , the entry  $(A^k)_{ij}$  equals the number of walks of length exactly  $k$  from node  $i$  to node  $j$ , where a walk is a sequence of nodes in which consecutive members are adjacent and repetition is permitted.

*Proof.* Induction on  $k$ . For  $k = 0$ ,  $A^0 = I$  and there is exactly one walk of length zero from  $i$  to itself and none to any other node. Assume the claim for  $k$ . Every walk of length  $k + 1$  from  $i$  to  $j$  decomposes uniquely into a walk of length  $k$  from  $i$  to some intermediate node  $\ell$  followed by a single step from  $\ell$  to  $j$ , and the step exists precisely when  $A_{\ell j} = 1$ . Summing over the possible  $\ell$ ,

$$\#\{\text{walks } i \rightarrow j \text{ of length } k+1\} = \sum_{\ell} (A^k)_{i\ell} A_{\ell j} = (A^{k+1})_{ij}. \quad \blacksquare$$

**Corollary 3** (Receptive field). Let  $\tilde{A} = A + I$ . After  $K$  layers of message passing, the hidden state  $h_i^{(K)}$  depends on the input features of node  $j$  if and only if  $(\tilde{A}^K)_{ij} > 0$ , that is if and only if the graph distance  $\text{dist}(i, j) \leq K$ .

**Corollary 4** (Spectral moments).  $\text{tr}(A^k) = \sum_{\ell} \lambda_{\ell}^k$  counts closed walks of length  $k$ . In particular  $\text{tr}(A^2) = 2m$  counts each bond in both directions, and  $\text{tr}(A^6)$  is sensitive to the presence of six-membered rings.

Corollary 4 is the bridge between a graph model and electronic structure. The density of states of a Hückel system is determined by the moments  $\text{tr}(A^k)$ , and a linear message-passing model with sum aggregation and readout computes precisely a weighted sum of such moments. Problem 12 makes the statement quantitative.

### 3 Permutation symmetry

Let  $P$  be an  $n \times n$  permutation matrix, so  $P_{ij} = 1$  exactly when the relabelling sends  $j$  to position  $i$ . Relabelling acts on the data by  $X \mapsto PX$  and  $A \mapsto PAP^{\top}$ .

**Theorem 5** (Equivariance of a message-passing layer). Let one layer be defined by

$$h'_i = \phi_{\text{upd}}\left(h_i, \bigoplus_{j \in \mathcal{N}(i)} \phi_{\text{msg}}(h_i, h_j, e_{ji})\right), \quad (3)$$

with  $\phi_{\text{msg}}$  and  $\phi_{\text{upd}}$  shared by all nodes and  $\bigoplus$  invariant to the order of its arguments. Then the layer is permutation equivariant.

*Proof.* Write  $\pi$  for the permutation with matrix  $P$ , so that after relabelling the node in position  $i$  carries the data formerly at  $\pi^{-1}(i)$ . Adjacency is preserved in the sense that  $j \in \mathcal{N}(i)$  in the original graph if and only if  $\pi(j) \in \mathcal{N}(\pi(i))$  in the relabelled one, because  $(PAP^{\top})_{\pi(i)\pi(j)} = A_{ij}$ . The message set collected at position  $\pi(i)$  is therefore

$$\{\phi_{\text{msg}}(h_i, h_j, e_{ji}) : j \in \mathcal{N}(i)\},$$

which is the same *set* as the one collected at  $i$  before relabelling, though possibly enumerated in a different order. Since  $\bigoplus$  is order-independent the two aggregates are equal, and since  $\phi_{\text{upd}}$  receives the same pair of arguments the outputs agree. Hence  $h'_{\pi(i)} = h'_i$ , which written in matrix form is  $F(PX, PAP^{\top}) = PF(X, A)$ . ■

**Corollary 6** (Invariance of the readout). If  $\hat{y} = \psi(\bigoplus_{i \in \mathcal{V}} h_i^{(K)})$  with  $\bigoplus$  order-independent, then  $\hat{y}$  is permutation invariant. Composition of equivariant layers is equivariant, and an invariant reduction of an equivariant object is invariant.

**Theorem 7** (Deep Sets representation, stated). Every continuous permutation-invariant function of a set of at most  $n$  elements drawn from a compact domain can be written in the form

$$f(\{x_1, \dots, x_n\}) = \psi\left(\sum_{i=1}^n \varphi(x_i)\right) \quad (4)$$

for suitable continuous  $\varphi$  and  $\psi$ .

The theorem is the licence for the entire architecture. It says that restricting the readout to the form “embed each atom, sum, then transform” costs no expressive power at all, provided the embedding is rich enough. The restriction that does cost power is not the sum but the locality of the message function, and Section 4 quantifies the loss.

**A remark on why the sum, and not the average, appears in Theorem 7.**

The representation requires an aggregator that retains the multiplicity of its arguments. An average discards the count, so the pair of multisets  $\{a\}$  and  $\{a, a\}$  collapse to the same value and no choice of  $\psi$  can separate them. Lemma 8 makes the point precise.

## 4 Injectivity on multisets and the limits of expressive power

A neighbourhood is a *multiset*, an unordered collection in which repetition matters. Two carbons with neighbour multisets  $\{H, H, C\}$  and  $\{H, H, H\}$  must not be confused, so the aggregator has to be injective on multisets.

**Lemma 8** (Sum is injective, mean and maximum are not). Let the elements of a multiset be drawn from a countable set  $\mathcal{X}$  and let the multiset have at most  $N$  elements. There exists  $\varphi : \mathcal{X} \rightarrow \mathbb{R}$  such that  $S \mapsto \sum_{x \in S} \varphi(x)$  is injective on such multisets. No choice of  $\varphi$  makes  $S \mapsto \frac{1}{|S|} \sum_{x \in S} \varphi(x)$  or  $S \mapsto \max_{x \in S} \varphi(x)$  injective.

*Proof.* Enumerate  $\mathcal{X} = \{x_1, x_2, \dots\}$  and set  $\varphi(x_t) = (N+1)^{-t}$ . A multiset  $S$  with multiplicities  $c_t \in \{0, \dots, N\}$  maps to  $\sum_t c_t (N+1)^{-t}$ , which is the base- $(N+1)$  expansion of a number with digits  $c_t$ . The expansion is unique, so the multiplicities and hence  $S$  are recoverable, and the map is injective.

For the mean, take  $S_1 = \{x\}$  and  $S_2 = \{x, x\}$ . Both give  $\varphi(x)$  for every  $\varphi$ , and the two multisets differ. For the maximum, take  $S_1 = \{x, y\}$  and  $S_2 = \{x, y, y\}$  with  $\varphi(y) > \varphi(x)$ , both giving  $\varphi(y)$ . ■

### Colour refinement, 1-WL

Set  $c_i^{(0)}$  equal to the initial label of node  $i$ . For  $k \geq 0$ ,

$$c_i^{(k+1)} = \text{HASH}\left(c_i^{(k)}, \{\{c_j^{(k)} : j \in \mathcal{N}(i)\}\}\right), \quad (5)$$

where  $\{\{\cdot\}\}$  denotes a multiset and HASH is injective. Two graphs are declared distinguishable if at some round the multisets of colours differ.

**Theorem 9** (Expressive upper bound, stated). If 1-WL assigns identical colour multisets to two graphs at every round, then every message-passing model of the form of Theorem 5 assigns them identical graph-level outputs. Conversely, a model whose  $\phi_{\text{msg}}$ ,  $\oplus$  and  $\phi_{\text{upd}}$  are jointly injective on multisets is as discriminative as 1-WL.

The forward direction is proved by induction on the layer index: if the colours agree at round  $k$  then the hidden states agree, because the update is a function of the same arguments. The converse is the content of the GIN construction, which realises injectivity with  $h'_i = \text{MLP}((1+\epsilon)h_i + \sum_j h_j)$  and Lemma 8.

**Corollary 10.** Any two connected  $d$ -regular graphs with identical node labels and the same number of nodes are indistinguishable by 1-WL, hence by any standard message-passing model.

*Proof.* Every node has the same initial colour and the same neighbour multiset of size  $d$ , so refinement reaches its fixed point at round one with a single colour class in both graphs. The colour multisets agree at every round. ■

Corollary 10 is the source of the cycle-blindness demonstrated in Problem 11: all cycloalkane carbon skeletons are 2-regular.

## 5 Where graph convolution comes from

The convolution of two functions on the line is a product in Fourier space. The graph analogue replaces the Fourier basis by the eigenbasis of the Laplacian.

Let  $L_{\text{sym}} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$  with orthonormal  $\mathbf{U}$  and  $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$ . The *graph Fourier transform* of a signal is  $\hat{f} = \mathbf{U}^\top f$ , and a spectral filter with response  $g$  acts as

$$f \longmapsto \mathbf{U} g(\mathbf{\Lambda}) \mathbf{U}^\top f = g(L_{\text{sym}})f. \quad (6)$$

Equation (6) is exact and completely impractical, since it needs the full eigendecomposition at cost  $O(n^3)$  and produces filters that do not transfer between graphs of different sizes. The remedy is to restrict  $g$  to a polynomial. Expanding in Chebyshev polynomials  $T_k$  on the rescaled variable  $\tilde{\lambda} = 2\lambda/\lambda_{\max} - 1$ ,

$$g(L_{\text{sym}}) \approx \sum_{k=0}^K \theta_k T_k(\tilde{L}), \quad (7)$$

which is now a polynomial of degree  $K$  in the Laplacian and therefore, by Theorem 2, strictly local: it mixes information over at most  $K$  bonds and costs  $O(Km)$ .

Take  $K = 1$  and  $\lambda_{\max} \approx 2$ , which is exact for a bipartite graph and a good approximation otherwise. With  $T_0 = 1$  and  $T_1(\tilde{\lambda}) = \tilde{\lambda}$ ,

$$g(L_{\text{sym}}) \approx \theta_0 I + \theta_1 (L_{\text{sym}} - I) = \theta_0 I - \theta_1 D^{-1/2} A D^{-1/2}. \quad (8)$$

Constraining  $\theta = \theta_0 = -\theta_1$  to reduce the parameter count gives

$$g(L_{\text{sym}}) \approx \theta (I + D^{-1/2} A D^{-1/2}). \quad (9)$$

The operator in brackets has eigenvalues in  $[0, 2]$ , so repeated application is numerically unstable. Replacing  $A$  by  $\tilde{A} = A + I$  and  $D$  by the corresponding  $\tilde{D}$ , a step known as the renormalisation trick, yields the familiar layer

$$H^{(k+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(k)} W^{(k)}). \quad (10)$$

Reading (10) entrywise recovers the neighbourhood average of Section 8 of the main tutorial. The spectral derivation and the message-passing picture are the same object approached from two sides, and the derivation explains where the factor  $1/\sqrt{d_i d_j}$  comes from rather than presenting it as a normalisation chosen by taste.

## 6 Oversmoothing as a contraction

**Theorem 11** (Decay of Dirichlet energy). Let  $P_{\text{rw}} = \tilde{D}^{-1} \tilde{A}$  be the random-walk operator with self-loops on a connected non-bipartite graph, and let  $\mu = \max\{|\lambda| : \lambda \in \text{spec}(P_{\text{rw}}), \lambda \neq 1\} < 1$ . For the linear propagation  $H^{(k+1)} = P_{\text{rw}} H^{(k)}$ ,

$$\|H^{(k)} - \mathbf{1} \pi^\top H^{(0)}\| \leq C \mu^k, \quad (11)$$

where  $\pi$  is the stationary distribution. Every row converges to the same vector at a geometric rate governed by  $\mu$ .

*Proof.*  $P_{\text{rw}}$  is a stochastic matrix, so  $\mathbf{1}$  is a right eigenvector with eigenvalue 1, and by Perron–Frobenius the eigenvalue is simple with all others of modulus at most  $\mu < 1$  under the stated conditions. Decomposing  $H^{(0)}$  in the eigenbasis, the component along  $\mathbf{1}$  is preserved while every other component is multiplied by its eigenvalue at each step, hence is bounded by  $\mu^k$  times its initial size. Summing the finitely many remaining components gives the constant  $C$ . ■

### Numerical illustration.

For the benzene skeleton with self-loops,  $P_{\text{rw}}$  has eigenvalue moduli 1, 0.667, 0.667, 0.333, 0, 0, so  $\mu = 2/3$ . Starting from random node features, the largest deviation of any row from the mean row falls as 1.080, 0.400, 0.252, 0.114, 0.023, 0.0009 at  $k = 0, 1, 2, 4, 8, 16$ . The ratio between successive even steps is close to  $\mu^2 = 0.444$ , exactly as the theorem predicts. After sixteen layers the six carbons are numerically identical, and any model attempting to distinguish them has failed.

Nonlinearity and a residual connection replace  $P_{\text{rw}}$  by  $\frac{1}{2}(I + P_{\text{rw}})$  in effect, which raises  $\mu$  towards one and slows the collapse without preventing it. The practical conclusion is unchanged: depth is bounded above by the smoothing rate of the graph, and molecular graphs, being small and well connected, smooth quickly.

## 7 Gradients through gather and scatter

Training requires the derivative of the loss with respect to every hidden state. Message passing is built from two adjoint operations, and knowing the pair is enough.

**Proposition 12.** Let  $a_i = \sum_{j \in \mathcal{N}(i)} m_{j \rightarrow i}$  with  $m_{j \rightarrow i} = \phi_{\text{msg}}(h_i, h_j, e_{ji})$ . Then

$$\frac{\partial \mathcal{L}}{\partial h_j} = \underbrace{\frac{\partial \mathcal{L}}{\partial h_j}}_{\text{own update}} + \sum_{i \in \mathcal{N}(j)} \left( \frac{\partial m_{j \rightarrow i}}{\partial h_j} \right)^\top \frac{\partial \mathcal{L}}{\partial a_i}. \quad (12)$$

In words, the forward pass gathers along edges and scatters into nodes, and the backward pass gathers along the reversed edges and scatters into the same nodes. The adjoint of a sum aggregation is a broadcast, and the adjoint of a broadcast is a sum. Memory cost is therefore  $O(md)$  for the stored messages, which for a molecule of a few tens of atoms is negligible and for a protein graph is the binding constraint.

### Chemist's note: why the gradient is cheap here

An atom influences the loss through every path of length at most  $K$  leaving it, and the number of such paths grows quickly. The backward pass nevertheless costs the same as the forward pass, because the shared intermediate quantities are computed once and reused, exactly as in the recursive evaluation of a matrix power. The situation is the discrete analogue of the  $2n + 1$  rule in perturbation theory, where derivatives come far more cheaply than a naive count of terms suggests.

## 8 Euclidean symmetry and why forces need equivariance

Let a conformer be a list of positions  $\{r_i\}$  and let  $R \in O(3)$ ,  $t \in \mathbb{R}^3$  act by  $r_i \mapsto Rr_i + t$ .

### Invariant and equivariant targets

A scalar  $E$  is invariant if  $E(\{Rr_i + t\}) = E(\{r_i\})$ . A vector field  $F_i$  is equivariant if  $F_i(\{Rr_j + t\}) = R F_i(\{r_j\})$ . A rank-two tensor  $T$  transforms as  $RTR^\top$ .

**Theorem 13** (Forces inherit equivariance from an invariant energy). If  $E$  is differentiable and invariant under  $O(3)$  and translations, then  $F_i = -\partial E / \partial r_i$  is equivariant, and in addition  $\sum_i F_i = 0$ .

*Proof.* Write  $E_R(\{r_j\}) = E(\{Rr_j\})$ . Invariance says  $E_R = E$  for every  $R$ . Differentiating the left side by the chain rule,

$$\frac{\partial}{\partial r_i} E(\{Rr_j\}) = R^\top (\nabla_i E)(\{Rr_j\}),$$

and differentiating the right side gives  $(\nabla_i E)(\{r_j\})$ . Equating and multiplying by  $R$ ,

$$(\nabla_i E)(\{Rr_j\}) = R (\nabla_i E)(\{r_j\}),$$

which is equivariance of the gradient and hence of  $F_i$ . For the sum rule, apply invariance under the translation  $r_j \mapsto r_j + t$  and differentiate with respect to  $t$  at  $t = 0$ , which gives  $\sum_i \nabla_i E = 0$ . ■

Theorem 13 explains why an architecture with scalar hidden features can still produce correct forces. It builds an invariant  $E$  from interatomic distances, which are themselves invariant, and differentiates. The price is that directional information is reconstructed at every evaluation rather than carried, and the backward pass is mandatory even at inference time.



An equivariant architecture instead carries features indexed by the irreducible representations of  $SO(3)$ . A feature of type  $\ell = 0$  is a scalar,  $\ell = 1$  a vector,  $\ell = 2$  a symmetric traceless tensor. Messages combine features by the tensor product with Clebsch–Gordan coefficients,

$$(h^{(\ell_1)} \otimes h^{(\ell_2)})_{m_3}^{(\ell_3)} = \sum_{m_1, m_2} C_{\ell_1 m_1 \ell_2 m_2}^{\ell_3 m_3} h_{m_1}^{(\ell_1)} h_{m_2}^{(\ell_2)}, \quad (13)$$

which is the same coupling rule used for angular momenta in spectroscopy. The selection rule  $|\ell_1 - \ell_2| \leq \ell_3 \leq \ell_1 + \ell_2$  is the familiar one, and a chemist who has coupled two  $p$  orbitals has already performed the calculation that an equivariant message performs.

#### Why distances alone are not enough.

The full set of interatomic distances determines a rigid structure up to reflection, so a distance-only model cannot distinguish enantiomers. Worse, a model that uses only distances *within a cutoff* sees strictly less, and there exist distinct local environments with identical two-body distance distributions. Angular terms remove most of the degeneracy and equivariant vector features remove it more cheaply, which is the content of the ladder in the main tutorial.

## Part B. Twelve chemical problems, worked

Each problem states a chemical question, solves it in full, and closes with the single point it is designed to teach. Problems 1 to 4 concern what a graph model can perceive, 5 to 8 concern architectural choices forced by chemistry, and 9 to 12 concern making a model behave correctly in practice.

### 9 Constitutional isomers: ethanol against dimethyl ether

#### Problem 1

Ethanol and dimethyl ether share the formula  $C_2H_6O$ . Working with heavy atoms only, so that ethanol is the path C–C–O and dimethyl ether the path C–O–C:

- Show that a model with no message passing, which embeds each atom independently and sums, assigns the two the same output.
- Run 1-WL colour refinement for two rounds on both and determine the first round at which they separate.
- State the minimum number of message-passing layers required.

(a) Without message passing the readout is  $\hat{y} = \psi(\sum_i \varphi(x_i))$ . Both structures contain the multiset of atom types  $\{C, C, O\}$ , so the sums are term-by-term identical for every  $\varphi$ , and the outputs coincide. The same argument applies to the bond multiset, which is  $\{\text{single}, \text{single}\}$  in both cases.

(b) Write  $c$  for carbon and  $o$  for oxygen at round 0.



|                             | atom 1           | atom 2              | atom 3           |
|-----------------------------|------------------|---------------------|------------------|
| <b>ethanol</b> C–C–O        |                  |                     |                  |
| round 0                     | $c$              | $c$                 | $o$              |
| round 1                     | $(c, \{\{c\}\})$ | $(c, \{\{c, o\}\})$ | $(o, \{\{c\}\})$ |
| relabelled                  | <b>1</b>         | <b>0</b>            | <b>2</b>         |
| <b>dimethyl ether</b> C–O–C |                  |                     |                  |
| round 0                     | $c$              | $o$                 | $c$              |
| round 1                     | $(c, \{\{o\}\})$ | $(o, \{\{c, c\}\})$ | $(c, \{\{o\}\})$ |
| relabelled                  | <b>0</b>         | <b>1</b>            | <b>0</b>         |

At round 0 the colour multisets agree, both being  $\{\{c, c, o\}\}$ . At round 1 ethanol gives three *distinct* colours while dimethyl ether gives only two, one of them with multiplicity two. The multisets differ, so the structures separate at round 1.

(c) One layer suffices. A single round of message passing reproduces the round-1 colouring, after which the sum readout sees three distinct atom representations in one case and a repeated one in the other.

### Teaching point.

Constitution is invisible to composition and becomes visible after exactly one exchange of information between bonded atoms. The step from a bag of atoms to a graph model is not incremental, it is the whole of structural chemistry.

## 10 Branching: *n*-butane against isobutane

### Problem 2

Consider the carbon skeletons of *n*-butane, a path of four carbons, and isobutane, a central carbon bonded to three others. All atoms carry the same initial label.

- Write the degree sequence of each.
- Run one round of colour refinement and confirm separation.
- Show that a model using *mean* aggregation with a linear message function and uniform input features cannot separate them, whereas *sum* aggregation can.

(a) *n*-Butane has degrees (1, 2, 2, 1) and isobutane (3, 1, 1, 1). Both have four nodes and three bonds, as every acyclic graph on four nodes must.

(b) Starting from a single colour 0, one round gives

$$n\text{-butane} : \{(0, \{\{0\}\}), (0, \{\{0, 0\}\}), (0, \{\{0, 0\}\}), (0, \{\{0\}\})\} \rightarrow \{\{a, b, b, a\}\},$$

$$\text{isobutane} : \{(0, \{\{0, 0, 0\}\}), (0, \{\{0\}\})^{\times 3}\} \rightarrow \{\{c, a, a, a\}\},$$

with *a* the colour of a degree-one carbon. The multisets contain two colours in the ratio 2:2 against 3:1, so refinement separates them after one round. The distinguishing quantity is precisely the degree.

(c) Set every input feature to 1. With mean aggregation the aggregate at node *i* is  $\frac{1}{d_i} \sum_{j \in \mathcal{N}(i)} 1 = 1$  for every node in every graph, so all hidden states are identical and the two readouts agree. With sum aggregation the aggregate is  $d_i$ , giving (1, 2, 2, 1) and (3, 1, 1, 1), whose sums agree at  $6 = 2m$  but whose subsequent nonlinear transforms do not. Applying any nonlinear  $\psi$  elementwise before the readout separates them, for example  $\sum_i d_i^2 = 10$  against 12.

**Teaching point.**

Degree is substitution pattern. Primary, secondary, tertiary and quaternary carbon are degrees one to four, and an aggregator that normalises degree away has discarded the classification on which most of organic chemistry is organised.

## 11 Cycle blindness and its cure

**Problem 3**

Compare a graph consisting of two disjoint six-membered carbon rings with a single twelve-membered ring, all atoms carbon, no substituents.

- Prove that 1-WL cannot distinguish them.
- Compute  $\text{tr}(A^2)$  and  $\text{tr}(A^6)$  for both and comment.
- Give two practical remedies, and say which is cheaper.

(a) Both graphs are 2-regular on twelve identically labelled nodes. By Corollary 10 refinement reaches its fixed point immediately with a single colour class, and the colour multisets are  $\{\{a\}\}^{12}$  in both cases at every round. No message-passing model of the standard form can separate them, whatever its width, depth or training data.

(b) By Corollary 4,  $\text{tr}(A^2) = 2m$ . Both graphs have twelve bonds, so both give 24. The sixth moment counts closed walks of length six, and here the two differ:

$$\text{tr}(A^6) = 2 \times 132 = 264 \text{ (two rings),} \quad \text{tr}(A^6) = 240 \text{ (one ring).}$$

The excess of 24 is exactly the contribution of walks that traverse a six-membered ring once, in either direction, starting from each of the twelve atoms. The adjacency spectra also differ, being  $\{\pm 2, \pm 1, \pm 1\}$  doubled for the two rings against  $\{\pm 2, \pm\sqrt{3}, \pm\sqrt{3}, \pm 1, \pm 1, 0, 0\}$  for the macrocycle.

(c) Either supply a node feature recording the size of the smallest ring containing the atom, which takes value 6 and 12 respectively and separates the graphs at the input layer, or supply spectral features derived from  $\text{tr}(A^k)$  or from the leading Laplacian eigenvectors. The ring-size feature is cheaper by a wide margin, since ring perception is a linear-time graph traversal available in every cheminformatics toolkit, whereas the spectrum costs  $O(n^3)$  and does not transfer cleanly between molecules of different size.

**Teaching point.**

The cure for an expressivity failure is almost always a better input feature, not a more elaborate architecture. Higher-order models that provably exceed 1-WL exist, and a ring-size integer beats them on cost by several orders of magnitude.

## 12 The Hückel bridge

**Problem 4**

For benzene in the Hückel approximation,  $H = \alpha I + \beta A$  with  $A$  the adjacency matrix of the six-membered carbon ring.

- Derive the eigenvalues of  $A$  for a cycle  $C_n$  in closed form and evaluate for  $n = 6$ .
- Obtain  $E_\pi$  for six  $\pi$  electrons and the delocalisation energy relative to three ethylene units.

(c) Show that a linear message-passing model with uniform input features and sum readout computes walk counts, and identify what such a model can and cannot learn about the Hückel spectrum.

(a) Index the ring atoms by  $t = 0, \dots, n-1$  with  $A_{t,t\pm 1} = 1$  modulo  $n$ . Try  $\psi_t^{(k)} = \omega^{kt}$  with  $\omega = e^{2\pi i/n}$ . Then

$$(A\psi^{(k)})_t = \omega^{k(t-1)} + \omega^{k(t+1)} = (\omega^{-k} + \omega^k)\omega^{kt} = 2\cos\left(\frac{2\pi k}{n}\right)\psi_t^{(k)},$$

so  $x_k = 2\cos(2\pi k/n)$  for  $k = 0, \dots, n-1$ . The eigenvector is a plane wave on the ring, which is the discrete analogue of a particle on a circle and the reason the result matches the free-electron model. For  $n = 6$ ,

$$x = 2, 1, 1, -1, -1, -2.$$

(b) Orbital energies are  $\varepsilon_k = \alpha + \beta x_k$  with  $\beta < 0$ , so the lowest three levels are  $x = 2, 1, 1$ . Filling each with two electrons,

$$E_\pi = 2[(\alpha + 2\beta) + (\alpha + \beta) + (\alpha + \beta)] = 6\alpha + 8\beta.$$

Three isolated ethylene units, each with  $x = \pm 1$ , give  $3 \times (2\alpha + 2\beta) = 6\alpha + 6\beta$ . The delocalisation energy is therefore  $2\beta$ , the classical Hückel result.

(c) Take  $\mathbf{h}^{(0)} = \mathbf{1}$  and the linear layer  $\mathbf{h}^{(k+1)} = \mathbf{A}\mathbf{h}^{(k)}$  with readout  $\hat{y} = \sum_{k=0}^K c_k \mathbf{1}^\top \mathbf{A}^k \mathbf{1}$ . By Theorem 2 the term  $\mathbf{1}^\top \mathbf{A}^k \mathbf{1}$  is the total number of walks of length  $k$  in the graph. For benzene the counts are

$$12, 24, 48, 96, 192, 384 \quad (k = 1, \dots, 6),$$

which is  $6 \cdot 2^k$ , since the ring is 2-regular and  $\mathbf{A}\mathbf{1} = 2\mathbf{1}$ . Uniform features therefore reveal only the degree, and the model learns nothing about ring size. The informative quantity is instead  $\text{tr}(\mathbf{A}^k) = \sum_\ell x_\ell^k$ , the moments of the Hückel spectrum, which requires per-node information rather than a globally uniform signal. A model retaining distinct node states does access those moments, and a depth- $K$  model accesses them up to order  $K$ . Since  $E_\pi$  is a functional of the spectrum, and the spectrum is determined by its moments, a graph model of sufficient depth can in principle reproduce Hückel energies. Reproducing them to a given accuracy requires as many moments as the density of states demands.

#### Teaching point.

A graph model is a moment expansion of an electronic structure problem, truncated at the depth of the model. The statement makes the depth hyperparameter physically meaningful and explains why extended conjugation demands more layers than a localised functional group does.

## 13 Coordination number at a catalyst surface

### Problem 5

On an fcc(111) metal surface the coordination numbers are 12 for a bulk atom, 9 for a terrace atom, 7 for a step-edge atom and 6 for a kink atom. Adsorption energy correlates strongly with the coordination number of the binding site.

(a) Build the graph by a distance cutoff and state what the degree of a node means

physically.

- (b) Show that a model with mean aggregation and constant input features assigns the same representation to terrace, step and kink atoms.
- (c) Propose a minimal fix that preserves the stability advantage of normalisation.

(a) With a cutoff placed between the first and second neighbour shells, an edge exists exactly when two atoms are nearest neighbours, so  $d_i$  is the coordination number of atom  $i$ . Degree is not an abstraction here, it is a measurable structural quantity that appears directly in  $d$ -band centre arguments.

(b) With  $h_j^{(0)} = \mathbf{1}$  for all  $j$ , mean aggregation gives  $\frac{1}{d_i} \sum_{j \in \mathcal{N}(i)} \mathbf{1} = \mathbf{1}$  independently of  $d_i$ . Every site, whatever its coordination, produces the same aggregate, and the update  $\phi_{\text{upd}}(\mathbf{1}, \mathbf{1})$  is a constant. Terrace, step and kink become indistinguishable, and the model cannot express the very trend it was built to predict. GCN normalisation  $1/\sqrt{d_i d_j}$  suffers a milder version of the same defect, since it retains only  $\sqrt{d_i}$  dependence through the self term.

(c) Two options, both cheap. Concatenate the degree, or a smooth function of it, to the node features so that coordination survives whatever the aggregator does. Alternatively use a degree-scaled aggregator of the form  $\mathbf{a}_i = (1 + \log(1 + d_i)) \cdot \text{mean}_j \mathbf{m}_{j \rightarrow i}$ , which keeps the variance control of the mean while restoring monotone degree sensitivity.

#### Teaching point.

The aggregator is not a neutral implementation detail. Choosing mean aggregation for a surface science problem deletes coordination number, which is the leading descriptor in the field.

## 14 Extensivity and the choice of readout

### Problem 6

Atomisation energies of the linear alkanes follow  $E(n) \approx an + b$  to good accuracy, with  $n$  the number of carbons. Take  $a = -1200$  and  $b = -300 \text{ kJ mol}^{-1}$  and a training set  $n = 1, \dots, 12$ .

- (a) A model with mean readout and no size information can at best output a constant. Compute that constant and the signed error at  $n = 1, 6$  and  $12$ .
- (b) Show that the error is linear in  $n$  and give its slope.
- (c) Explain in one line why sum readout removes the defect.

(a) The best constant under squared loss is the mean of the targets. With  $\bar{n} = 6.5$ ,

$$\bar{E} = a\bar{n} + b = -1200(6.5) - 300 = -8100 \text{ kJ mol}^{-1}.$$

The signed errors  $\bar{E} - E(n)$  are

$$n = 1 : -6600, \quad n = 6 : -600, \quad n = 12 : +6600 \text{ kJ mol}^{-1}.$$

(b)  $\bar{E} - E(n) = a(\bar{n} - n)$ , which is linear in  $n$  with slope  $-a = +1200 \text{ kJ mol}^{-1}$  per carbon. The error changes sign at  $n = \bar{n}$ , giving the characteristic straight line through the mean training size that identifies the fault on sight.

(c) Sum readout produces an output proportional to the number of atoms by construction, so a model that learns a per-atom contribution reproduces an extensive target exactly, and the residual contains only the chemistry that deviates from additivity.

#### Teaching point.

Plot the signed error against molecular size before anything else. A straight line through the mean training size is diagnostic of a readout mismatch and takes one minute to find, whereas hyperparameter searching around the same fault can consume a week.

## 15 Stereochemistry: what the bond graph cannot hold

### Problem 7

A tetrahedral carbon bears four distinct substituents. Its two enantiomers have identical bond tables.

- Prove that no message-passing model acting on the bond graph alone can assign them different outputs.
- The two hydrogens of formaldehyde also received identical hidden states in the main tutorial. Explain why one case is a defect and the other is correct.
- Construct a feature that distinguishes the enantiomers and verify that it is invariant to atom relabelling.

(a) The two enantiomers have adjacency matrices related by a relabelling, that is  $A_2 = PA_1P^\top$  with the node features permuted identically. By Theorem 5 the layers are equivariant and by its corollary the readout is invariant, so the graph-level outputs are equal. The conclusion is independent of training.

(b) The formaldehyde hydrogens are related by an automorphism of the labelled graph, so they are genuinely equivalent and any physical per-atom property must take the same value on both. Assigning them equal representations is correct behaviour. The enantiomers are *not* related by any symmetry of the physical system, since a chiral environment distinguishes them, yet they are related by a symmetry of the bond graph. The defect lies in the representation, which has discarded the information that separates them.

(c) Order the four substituents by Cahn–Ingold–Prelog priority as  $1 > 2 > 3 > 4$  and form the signed volume

$$\chi = \text{sign} \left[ (r_1 - r_4) \cdot ((r_2 - r_4) \times (r_3 - r_4)) \right] \in \{+1, -1\}.$$

Attach  $\chi$  as a feature of the stereocentre. Relabelling the atoms does not change  $\chi$ , because the priority order is determined by substituent structure and not by index, so the feature is permutation invariant while remaining sensitive to reflection. Note also that  $\chi$  changes sign under an improper rotation, correctly encoding that the property is  $SO(3)$  invariant but not  $O(3)$  invariant.

#### Teaching point.

Identical representations for symmetry-equivalent atoms is the architecture working. Identical representations for enantiomers is the featurisation failing. Distinguishing the two situations is a question about the chemistry, not about the model.

## 16 Choosing depth from a Hammett argument

### Problem 8

A substituent  $X$  is placed *para* to a carboxylic acid on a benzene ring, and the target is the acid dissociation constant.

- (a) Count the bonds between the substituent atom and the acidic proton.
- (b) State the minimum depth  $K$  for the substituent to influence the prediction at the acidic site.
- (c) A colleague proposes  $K = 8$  for safety. Give the argument against, with reference to Theorem 11.

(a) Label the ring carbon bearing  $X$  as  $C_1$  and the ring carbon bearing the acid as  $C_4$ . The shortest path runs  $X \rightarrow C_1 \rightarrow C_2 \rightarrow C_3 \rightarrow C_4 \rightarrow \text{C(O)OH carbon} \rightarrow \text{O} \rightarrow \text{H}$ , giving seven bonds from the substituent atom to the acidic hydrogen and four bonds from  $X$  to  $C_4$ .

(b) If the readout aggregates over all atoms, information must reach the readout rather than any particular atom, and  $K = 4$  suffices for the substituent to reach the ring carbon bearing the acid. If instead the prediction is made at the acidic hydrogen as a node-level task,  $K \geq 7$  is required. The distinction is worth making explicit to students, because the two designs differ by nearly a factor of two in depth.

(c) At  $K = 8$  on a graph of roughly a dozen heavy atoms, the receptive field covers the entire molecule several times over, so the additional layers buy no new information. Theorem 11 then applies with a smoothing rate  $\mu$  close to that computed for benzene,  $\mu \approx 2/3$ , so the deviation of any atom from the molecular mean falls by a factor of about  $(2/3)^8 \approx 0.04$ . The ring carbons become numerically indistinguishable, the substituent effect is erased, and accuracy degrades. The correct move is to use the smallest  $K$  that covers the interaction range, with residual connections if a larger  $K$  is genuinely needed.

### Teaching point.

Depth is a chemical parameter. Count the bonds over which the effect operates, then set  $K$ , rather than treating depth as a quantity to be searched blindly.

## 17 Partial charges: a constrained node-level task

### Problem 9

A model predicts a partial charge  $q_i$  for each atom. Physically the charges must sum to the total molecular charge  $Q$ .

- (a) Show that an unconstrained node-level model does not satisfy the constraint.
- (b) Construct a correction that enforces it exactly, and prove that the corrected output is still permutation equivariant.
- (c) Apply the correction to the raw predictions (0.31, -0.44, 0.09, 0.10) for a neutral molecule.

(a) The model computes  $q_i = \psi(h_i^{(K)})$  independently for each atom. Nothing couples the outputs, so  $\sum_i q_i$  is whatever the parameters happen to produce, and the constraint holds only by accident.

(b) Define the projected charges

$$q_i^{\text{corr}} = q_i + \frac{1}{n} \left( Q - \sum_j q_j \right). \quad (14)$$

Summing gives  $\sum_i q_i^{\text{corr}} = \sum_i q_i + (Q - \sum_j q_j) = Q$  exactly. For equivariance, note that  $\sum_j q_j$  and  $n$  are both permutation invariant, so the correction adds the same scalar to every entry. If the atoms are relabelled by  $P$  then  $q \mapsto Pq$  by equivariance of the underlying model, and  $Pq + c\mathbf{1} = P(q + c\mathbf{1})$  because  $P\mathbf{1} = \mathbf{1}$ . Hence the corrected vector is equivariant.

(c) The raw sum is  $0.31 - 0.44 + 0.09 + 0.10 = 0.06$ . With  $Q = 0$  and  $n = 4$  the shift is  $-0.06/4 = -0.015$ , giving

$$(0.295, -0.455, 0.075, 0.085), \quad \sum_i q_i^{\text{corr}} = 0.000.$$

### Teaching point.

Physical constraints are imposed by construction, not learned. A projection layer costs nothing, holds exactly, and keeps the symmetry properties of the model intact. The same pattern enforces conservation of mass in a reaction model and zero net force in a potential.

## 18 Forces from a learned energy

### Problem 10

A model learns the energy of a diatomic as a function of the internuclear separation, fitted to a Morse form  $E(r) = D_e(1 - e^{-a(r-r_e)})^2$  with  $D_e = 4.75$  eV,  $a = 1.94$  Å<sup>-1</sup>,  $r_e = 0.741$  Å.

- Derive  $F_1$  and  $F_2$  in vector form and verify  $F_1 + F_2 = \mathbf{0}$ .
- Prove that the forces so obtained are rotationally equivariant.
- Evaluate the force magnitude at  $r = 0.900$  Å and state its direction.

(a) Write  $r = \|r_2 - r_1\|$  and  $\hat{u} = (r_2 - r_1)/r$ . Since  $\partial r / \partial r_2 = \hat{u}$  and  $\partial r / \partial r_1 = -\hat{u}$ ,

$$F_2 = -\frac{\partial E}{\partial r_2} = -E'(r) \hat{u}, \quad F_1 = +E'(r) \hat{u},$$

so the sum vanishes identically, which is Newton's third law recovered as a consequence of translational invariance, in accordance with Theorem 13.

(b) Under  $r_i \mapsto Rr_i$  the separation  $r$  is unchanged, since  $R$  is orthogonal, and the unit vector transforms as  $\hat{u} \mapsto R\hat{u}$ . Therefore  $F_2 \mapsto -E'(r)R\hat{u} = RF_2$ , which is exactly equivariance. The scalar  $E'(r)$  carries no direction and the direction is supplied entirely by  $\hat{u}$ , which is the mechanism by which an invariant model produces an equivariant output.

(c) Differentiating,  $E'(r) = 2D_e a(1 - e^{-x})e^{-x}$  with  $x = a(r - r_e)$ . At  $r = 0.900$  Å,  $x = 1.94 \times 0.159 = 0.3085$  and  $e^{-x} = 0.7345$ , so

$$E'(r) = 2(4.75)(1.94)(0.2655)(0.7345) = 3.593 \text{ eV }^{-1}.$$

The force on atom 2 is  $-3.593 \hat{u} \text{ eV }^{-1}$ , directed towards atom 1. The bond is stretched beyond  $r_e$ , so the force is restoring, as it must be.



**Teaching point.**

Predicting forces directly as three numbers per atom breaks equivariance unless the architecture is built for it, and it also breaks energy conservation, since an arbitrary vector field is not a gradient. Differentiating a learned energy gives both properties for free.

## 19 Batching arithmetic

**Problem 11**

Three molecules with 4, 3 and 5 atoms are batched. Their edge lists, in local indexing and with both directions stored, are

mol 1 : (0–1), (0–2), (0–3);   mol 2 : (0–1), (1–2);   mol 3 : (0–1), (1–2), (2–3), (3–4).

- Give the total node and directed edge counts and the `batch` vector.
- Write the offset edge list of the batch.
- With node states  $h_i = i$  for  $i = 0, \dots, 11$  in batch indexing, compute the sum readout of each molecule.

(a) Nodes:  $4 + 3 + 5 = 12$ . Directed edges:  $2(3 + 2 + 4) = 18$ . The batch vector assigns each atom to its molecule,

$$\text{batch} = [0, 0, 0, 0, 1, 1, 1, 2, 2, 2, 2, 2].$$

(b) Offsets are the running node counts 0, 4 and 7. Adding them to the local indices,

$$\text{src} = [0, 1, 0, 2, 0, 3, 4, 5, 5, 6, 7, 8, 8, 9, 9, 10, 10, 11],$$

$$\text{dst} = [1, 0, 2, 0, 3, 0, 5, 4, 6, 5, 8, 7, 9, 8, 10, 9, 11, 10].$$

No entry of `src` and `dst` pairs an atom of one molecule with an atom of another, so the batch adjacency is block diagonal and no message crosses between molecules.

(c) Segment sums over the batch vector,

$$0+1+2+3 = 6, \quad 4+5+6 = 15, \quad 7+8+9+10+11 = 45.$$

**Teaching point.**

Variable-size inputs need no padding and no masking. The disjoint union turns a ragged batch into one graph, and the only bookkeeping is an index offset. Getting the offset wrong produces messages between unrelated molecules, which trains without error and predicts nonsense.

## 20 Designing the split

**Problem 12**

A dataset contains 2000 measured solubilities. Inspection shows that 600 of the entries are members of four congeneric series, each series differing only in the identity of one substituent.

- Estimate the optimistic bias introduced by a random 80:20 split.
- Describe the scaffold split that should be used instead.
- State what the two reported errors together tell a reader, and why both are worth

reporting.

(a) Under a random split each series is divided in the same proportion, so roughly 80% of the members of every series appear in training. A test compound drawn from a series therefore has close analogues in training, differing at a single position, and the task reduces to interpolating a substituent effect within a known framework. Around 30% of the test set, being the fraction from congeneric series, is affected. The reported error will be substantially lower than the error on a new framework, and the gap is often a factor of two or more in root-mean-square error on solubility data.

(b) Compute the Bemis–Murcko framework of each molecule, that is the ring systems together with the linkers joining them, with all side chains removed. Group molecules by framework and assign whole groups to training or test, largest groups first, until the desired proportion is reached. No framework appears on both sides, so a test molecule never has a training analogue that differs only in substitution.

(c) The random-split error answers “how accurate is the model on chemistry resembling its training set?”, which is the relevant question for optimising within a known series. The scaffold-split error answers “how accurate is the model on a framework it has never seen?”, which is the relevant question for a discovery campaign. Reporting both, with the gap made explicit, tells the reader how much of the performance is interpolation and how much is generalisation. Reporting only the first is the most common form of optimism in the molecular machine learning literature.

#### Teaching point.

The split encodes the question being asked. Choosing it is a scientific decision of the same weight as choosing the architecture, and it should be made and justified before any model is trained.

## A Verification script

Every numerical value quoted in Part A and Part B is produced by the following script, which depends only on NumPy. Students should run it and confirm each number as they work through the problems.

```
1  """
2  huckel_wl_checks.py
3  Numerical verification of every quantity quoted in the companion volume.
4  """
5  import numpy as np
6  from itertools import combinations
7  np.set_printoptions(precision=4, suppress=True)
8
9  def cycle_adj(n):
10     A = np.zeros((n, n))
11     for i in range(n):
12         A[i, (i + 1) % n] = A[(i + 1) % n, i] = 1.0
13     return A
14
15  print("=== B4 Huckel / adjacency spectra ===")
16  A6 = cycle_adj(6)
17  ev6 = np.sort(np.linalg.eigvalsh(A6))[:-1]
18  print("C6 eigenvalues x_k :", ev6)
19  print("closed formula 2cos(2pi k/6):", np.sort([2*np.cos(2*np.pi*k/6) for k in
20     range(6)]))[:-1])
20  # pi energy with 6 electrons, E = sum over occupied of (alpha + x beta), two per
    level
```

```

21 occ = ev6[:3]
22 print("E_pi = 6a + %.1f b" % (2*occ.sum()))
23 print("3 x ethylene = 6a + 6.0 b    -> delocalisation = %.1f b" % (2*occ.sum() -
    6))
24
25 print("\n=== B3  two C6 versus one C12 ===")
26 A12 = cycle_adj(12)
27 ev12 = np.sort(np.linalg.eigvalsh(A12))[:-1]
28 ev66 = np.sort(np.concatenate([ev6, ev6]))[:-1]
29 print("C12 spectrum :", ev12)
30 print("2 x C6          :", ev66)
31 print("spectra identical? ", np.allclose(ev12, ev66))
32 print("tr(A^2)  C12 = %.0f    2xC6 = %.0f" % (np.trace(A12@A12), 2*np.trace(A6@A6
    )))
33 print("tr(A^6)  C12 = %.0f    2xC6 = %.0f" % (np.trace(np.linalg.matrix_power(A12
    ,6)),
    2*np.trace(np.linalg.matrix_power(
34     A6,6))))
35
36 def wl(adj, colours, rounds=3):
37     """1-WL colour refinement. Returns colour vector after each round."""
38     n = len(colours)
39     hist = [list(colours)]
40     cur = list(colours)
41     for _ in range(rounds):
42         sig = [(cur[i], tuple(sorted(cur[j] for j in range(n) if adj[i, j])))]
43         for i in range(n):
44             table, new = {}, []
45             for s in sorted(set(sig), key=str):
46                 table[s] = len(table)
47             new = [table[s] for s in sig]
48             if sorted(new) == sorted(cur) and len(set(new)) == len(set(cur)):
49                 hist.append(new); break
50             cur = new
51             hist.append(list(cur))
52     return hist
53
54 print("\n=== B1  ethanol versus dimethyl ether (heavy atoms only) ===")
55 # ethanol  C-C-O ; dimethyl ether C-O-C
56 Aeth = np.array([[0,1,0],[1,0,1],[0,1,0]]) # C0-C1-O2
57 Adme = np.array([[0,1,0],[1,0,1],[0,1,0]]) # C0-O1-C2 (same shape, different
    labels)
58 col_eth = [0,0,1] # C,C,O
59 col_dme = [0,1,0] # C,O,C
60 print("ethanol  colours:", wl(Aeth, col_eth, 2))
61 print("dme      colours:", wl(Adme, col_dme, 2))
62
63 print("\n=== B2  n-butane versus isobutane (carbon skeleton) ===")
64 Anb = np.array([[0,1,0,0],[1,0,1,0],[0,1,0,1],[0,0,1,0]])
65 Aib = np.array([[0,1,1,1],[1,0,0,0],[1,0,0,0],[1,0,0,0]])
66 print("n-butane degrees :", Anb.sum(1))
67 print("isobutane degrees:", Aib.sum(1))
68 print("n-butane WL  :", wl(Anb, [0,0,0,0], 2))
69 print("isobutane WL :", wl(Aib, [0,0,0,0], 2))
70
71 print("\n=== B6  mean readout on an extensive target ===")
72 # atomisation energy of linear alkanes CnH2n+2 modelled as E = a*n + b
73 a, b = -1200.0, -300.0 # kJ/mol, illustrative
74 ns = np.arange(1, 13)
75 E = a*ns + b
76 print("true E(n) = %.0f n %.0f" % (a, b))
77 print("best constant predictor (mean readout, size-blind) = %.0f" % E.mean())
78 err = E.mean() - E
79 print("signed error at n=1,6,12 :", err[[0,5,11]])

```

```

79 print("slope of the error with n :", np.polyfit(ns, err, 1)[0])
80
81 print("\n=== B9 charge conservation correction ===")
82 q_raw = np.array([0.31, -0.44, 0.09, 0.10]) # raw per-atom predictions,
      molecule neutral
83 Q = 0.0
84 q_fix = q_raw + (Q - q_raw.sum())/len(q_raw)
85 print("raw sum = %.3f" % q_raw.sum())
86 print("corrected =", q_fix, " sum = %.3f" % q_fix.sum())
87
88 print("\n=== B11 batching offsets ===")
89 sizes = [4, 3, 5]
90 edges = [np.array([0,1,0,2,0,3],[1,0,2,0,3,0]),
91          np.array([0,1,1,2],[1,0,2,1]),
92          np.array([0,1,1,2,2,3,3,4],[1,0,2,1,3,2,4,3])]
93 off, cols = 0, []
94 for s, e in zip(sizes, edges):
95     cols.append(e + off); off += s
96 big = np.concatenate(cols, axis=1)
97 batch = np.concatenate([[k]*s for k, s in enumerate(sizes)])
98 print("total nodes =", off, " total directed edges =", big.shape[1])
99 print("batch vector:", batch)
100 print("offset edge_index:\n", big)
101 h = np.arange(off, dtype=float)[: , None]
102 seg = np.zeros((3,1)); np.add.at(seg, batch, h)
103 print("segment sums:", seg.ravel())
104
105 print("\n=== A6 oversmoothing rate ===")
106 Ahat = cycle_adj(6) + np.eye(6)
107 Dinv = np.diag(1/Ahat.sum(1))
108 P = Dinv @ Ahat
109 ev = np.sort(np.abs(np.linalg.eigvals(P)))[::-1]
110 print("|eigenvalues| of row-normalised A-hat:", ev)
111 print("second largest modulus (smoothing rate) = %.4f" % ev[1])
112 H = np.random.default_rng(1).normal(size=(6,3))
113 for k in [0,1,2,4,8,16,32]:
114     Hk = np.linalg.matrix_power(P, k) @ H
115     spread = np.abs(Hk - Hk.mean(0)).max()
116     print(" k=%2d max deviation from the mean row = %.5f" % (k, spread))

```

**Listing 1:** *huckel\_wl\_checks.py*. Spectra, walk counts, colour refinement, readout error, charge projection, batching offsets and the oversmoothing rate.

## B Summary of results

| Result               | Where   | Statement                                                   |
|----------------------|---------|-------------------------------------------------------------|
| Walk counting        | Thm 2   | $(A^k)_{ij}$ counts walks of length $k$                     |
| Receptive field      | Cor 3   | depth $K$ sees exactly the ball of radius $K$               |
| Equivariance         | Thm 5   | shared $\phi$ plus order-free $\oplus$ implies equivariance |
| Deep Sets            | Thm 7   | embed, sum, transform loses no expressive power             |
| Multiset injectivity | Lem 8   | sum is injective, mean and maximum are not                  |
| Expressive bound     | Thm 9   | message passing is at most as strong as 1-WL                |
| Regular graphs       | Cor 10  | all $d$ -regular labelled graphs are confusable             |
| Graph convolution    | Eq (10) | first-order Chebyshev filter plus renormalisation           |
| Oversmoothing        | Thm 11  | deviation decays as $\mu^k$ , $\mu$ the spectral gap        |
| Forces               | Thm 13  | invariant energy implies equivariant, balanced forces       |

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End of companion volume